

Work Package — D_{IV}^5 as an invertible Berezin–Toeplitz engine: the projection has an inverse on the physical subspace, and the interior carries finite discrete math

Keeper | 2026-07-15 | Filed at Casey’s request. Captures a four-leap conceptual arc from the EOD conversation as one sequenced work package. Phase-two of the program (the substrate as a *calculational engine*, not just a parameter table). Honestly tiered: the mathematical anchors are real and precedented; NONE of it is demonstrated. Sequenced AFTER the mixing matrix lands (which lets the operator be written down).

The one-line idea (Casey)

Projections can be inverted — and the interior of D_{IV}^5 carries finite discrete math. So the Standard Model is (candidate) one finite linear operator on a discrete basis: forward = bulk reconstruction, inverse = quantization, and its matrix elements are the observables.

The arc, as four leaps (each with anchor + honest tier)

1. The transfer / projection operator (spectrum = masses, eigenvectors = mixing)

- The Bergman/Szegő projection on D_{IV}^5 is a genuine projection operator, not a metaphor. Discretized on the K-type basis it is a matrix. The dynamical object is its Toeplitz compression $P \cdot (\text{shift/dilation}) \cdot P$ (a pure projection is idempotent $\rightarrow$ carries no dynamics alone; the dynamics live in the compression). This is the operator class Lyra’s bulk-color work already uses ($\text{su}(3)$ Cartan–Weyl on the Hardy-space Toeplitz algebra).
- Its eigenvalues are the masses (our Σ); its eigenvectors are the mixing (U, V). This is the textbook lattice-QFT statement (transfer-matrix eigenvalues = the spectrum). So the static “physics matrix” $M = U \Sigma V^\dagger$ and the dynamical “step operator” are ONE object seen two ways.
- Anchor: transfer-matrix / Koopman methodology; lattice-QFT spectrum. Precedented $\rightarrow$ not new physics as a method (solid ground).
- Tier: BANKED spectrum (masses done), mixing PENDING (tomorrow). The operator itself is NOT written down.

2. The inverse projection = quantization (Berezin–Toeplitz)

- A projection is NOT invertible on the whole space (it has a kernel — the modes it annihilates). It IS a clean two-way isomorphism on the physical (holomorphic) subspace: Extension E (discrete boundary $\rightarrow$ 3D bulk continuum) and Restriction R (continuum $\rightarrow$ discrete), with $R \circ E = \text{id}$ on the subspace and $E \circ R = P$ on the bulk. The inverse exists exactly where the physics lives; off it, the projection has thrown information away (consistent with Casey’s own “limits are lossy, integrals preserve” — the Bergman projection is an integral against the reproducing kernel, hence preserving).
- The name for “continuum classical data $\rightarrow$ discrete quantum operator via projection onto holomorphic sections of a Kähler domain” is Berezin–Toeplitz geometric quantization on a bounded symmetric domain (Berezin, Rawnsley, Bordemann–Meinrenken–Schlichenmaier). D_{IV}^5 is Hermitian symmetric $\rightarrow$ exactly that setting.
- Two payoffs from the identification: (a) the “recursion” is literal — a reproducing-kernel space regenerates its own elements, $f(z) = \langle f, K_z \rangle$, so the RKHS self-reference IS the continuum $\leftrightarrow$ discrete tower Casey senses; (b) our weight $E_0 (=2)$, the FK weight that sets the masses) is the quantization parameter (the role of $1/\hbar$; classical/continuum limit = weight $\rightarrow \infty$). The mass-scale parameter and the classical $\leftrightarrow$ quantum knob are the same number.
- Novel wrinkle worth chasing: ordinary geometric quantization leaves $\hbar$ a free family ($\hbar \rightarrow 0$). Ours is pinned by the geometry ($E_0=2$ from spin, finite capacity 137). A quantization with NO free $\hbar$ is genuinely unusual.
- Tier: method PRECEDENTED (rigorous field — go stand on it); “this domain quantizes to the SM” + “ $\hbar$ pinned” is the OPEN BST content. Extension/restriction operators NOT written as matrices yet.

3. Finite discrete interior → loops become finite sums (linear algebra, no renormalization)

- A loop integral is a sum over the complete intermediate basis; the continuum's $\int d^4k$ and its UV divergences come ENTIRELY from “infinitely many modes at every point.” A finite mode count (the 137 capacity) makes that sum finite → a tensor contraction / trace over a discrete index. UV divergences never appear; renormalization is simply not needed. Every quantum correction becomes a finite, well-defined number.
- Berezin–Toeplitz makes it rigorous: the loop expansion reorganizes into the $1/N$ star-product expansion, each term algebraic; if the operator is genuinely finite-rank in a sector, the exact answer is a finite contraction (no expansion).
- Three honest edges: (a) loops don't vanish — the quantum-correction *physics* stays, now as finite sums; (b) finite $\neq$ small — the multi-particle (Fock) state space grows combinatorially, so it is *conceptually* simpler (no renormalization) but only *computationally* cheap at low order/few-body (the lattice-QCD lesson: finite but still a supercomputer); (c) undemonstrated — not one substrate amplitude has been computed and matched.
- Tier: conceptual claim (finite → linear algebra, no renormalization) CORRECT and important; computational-simplicity claim carries the finite $\neq$ small asterisk; whole thing ASPIRATIONAL until one finite loop is computed and matched.

4. THE VALIDITY CONDITION (Casey — the Keeper-shaped one)

“Linear math defeats analysis unless you have an identity which appears on the discrete side too.” - A discretization is exact only for structures with an exact discrete counterpart; anything that lives only in the limit is silently lost. This is the theorem behind every successful discretization: lattice gauge theory keeps gauge invariance exact (Wilson); Ginsparg–Wilson/overlap fermions supply a lattice chiral symmetry where naive lattices fail (Nielsen–Ninomiya doublers); symplectic integrators conserve an exact shadow Hamiltonian. - Why BST is well-placed (the precise reason): BST's discreteness is NOT a lattice (which breaks Lorentz and loses identities); it is a discrete spectrum (K-type ladder, integer weights) carrying a genuine unitary representation of $SO(5,2)$. Representation theory carries a continuous group's identities onto a discrete basis *exactly* — Casimir eigenvalues, branching/recurrence relations, the reproducing kernel. So every symmetry-descended identity (Ward, conservation laws) is guaranteed on both sides. The identities aren't discretized; they're *represented*. NOTE: the space is infinite-dimensional (holomorphic functions); the discreteness is the discrete spectrum + finite capacity in specific sectors, not finite-dimensionality. - The at-risk identities + the first concrete test: not every needed identity is symmetry-generated. Safe = representation-theoretic; at-risk = genuine-limit identities, and the sharpest is the anomaly (a classical symmetry that fails quantum-mechanically). The SM has the chiral anomaly ($\pi^0 \rightarrow \gamma\gamma$, measured). First test of the whole arc: does the discrete substrate reproduce the chiral anomaly exactly? If yes, the representation is faithful where it counts and the engine is trustworthy; if no, we've found a limit-only identity and located exactly where linear algebra is defeated.

Sequencing (do NOT start before the gate)

1. GATE: the mixing matrix lands (tomorrow — refraction rotation → Cabibbo, then $d=5 \rightarrow d=4$ + Majorana locus → PMNS). This gives the operator's eigenvectors, i.e., most of the eigen-decomposition of the candidate operator.
2. Write the operator down — the Toeplitz compression on the K-type basis; forward (E) and inverse-on-subspace (R) as explicit matrices; identify E_0 as the Berezin $\hbar$.
3. Ranging shot (cheapest falsifiable move): do NOT aim at a full cross section. Take the simplest *dynamical* quantity where messy dynamics cancels and geometry remains — a decay-width or cross-section RATIO — compute it as a ratio of substrate matrix elements, compare to data. ONE success turns the arc from “beautiful and probably right” into “demonstrated.”
4. Validity audit: classify each continuum identity the physics needs as representation-theoretic (safe) or limit-only (at-risk); run the chiral-anomaly reproduction test first.

Discipline flags (Cal #27 / Five-Absence / target-innocence)

- This is the prettiest lane on the board; the discipline fires hardest here. Nothing above is banked. Tiers are anchor-vs-delivery honest.
- Five-Absence CLEAN: every claim is about *calculational method/precision*, not new phenomena — no new particles. Consistent with the program’s character (and with no-proton-decay = baryon number robustly conserved, filed separately).
- The payoff is a *conditional* vision: exact finite calculations → sharper predictions → experiment design (“substrate engineering”). Coherent, logically sound, conditional on the engine existing. Tier ASPIRATIONAL until step 3 lands.

— Keeper, 2026-07-15. Filed at Casey’s request. The substrate as a finite, invertible, discrete linear-algebra engine — mass=spectrum, mixing=eigenvectors, quantization=inverse projection, loops=finite sums — VALID exactly for identities that appear on the discrete side (rep-theoretic; anomaly = first test). Gated behind the mixing matrix. Not banked; phase-two container for four EOD leaps. See [[Keeper_K704...]] (mixing = distinct object), [[Keeper_K701...]] ($E_0=2 = \text{the } \hbar$), [[Keeper_K699...]] (finite capacity / 4π).